



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 10 (Summary and Tutorial)

Topic: Functions

Summary for Functions

Relations and Functions

- The set of inputs of a function is known as the **domain** while the corresponding set of outputs is known as the **range**.
- To define a function completely, both the rule and domain must be specified.
- The range of a function can be obtained by looking at its graph (y -values).
- If a function is such that no two elements in the given domain have the same output, it is a **one-one function**.
- To check if a function is one-one, we use horizontal lines to check.

If function is one-one, such as $f : x \mapsto x^2, x > 2$, we write since **every** horizontal line $y = k, k \in [4, \infty)$, cuts the graph of f at one and only one point, f is a one-one function. Note that $[4, \infty)$ is the range of f .

If function is NOT one-one, such as $f : x \mapsto x^2, x \in \mathbb{R}$, we choose a specific line $y = k$ where k is a value in range of f . In this case, we can write since the horizontal line $y = 1$ cuts the graph of f at more than one point, f is not a one-one function.

Inverse Functions

- Given a function f , the **inverse function** f^{-1} exists if and only if f is one-one.
- To define f^{-1} , you must find the rule and domain.
- To find the rule of f^{-1} , let $y = f(x)$ and **make x the subject**, i.e. $x = f^{-1}(y)$.

Example to find the rule of inverse of $g : x \mapsto 1 + e^{-x}, x \in \mathbb{R}$

Let $y = 1 + e^{-x}$ and make x the subject to get $x = -\ln(y - 1)$.

- To find domain of f^{-1} , use the result: $D_{f^{-1}} = R_f$

Note also that $R_{f^{-1}} = D_f$.

- The graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of each other in the line $y = x$.

Composite Functions

- Given 2 functions f and g , the **composite function** gf exists if and only if $R_f \subseteq D_g$.
- To define gf , you must find the rule and domain.
- To find the rule of gf , find $f(x)$ first, then find $g(f(x))$ by replacing every x in the rule of g with $f(x)$.
- To find domain of gf , use the result: $D_{gf} = D_f$
- In general, $fg \neq gf$. i.e. composition of functions is not commutative.
- The composite function ff , if it exists, is written as f^2 , i.e. $f^2(x) = ff(x) = f(f(x))$.
 - Useful results: (i) $ff^{-1}(x) = x$, where $x \in D_{f^{-1}}$.
(ii) $f^{-1}f(x) = x$, where $x \in D_f$.
- To find the range of gf , use either
 - (1) $R_{gf} = R_g$ restricted to R_f**or**
 - (2) sketch the graph of $y = gf(x)$ over D_{gf} and find R_{gf} .

Example

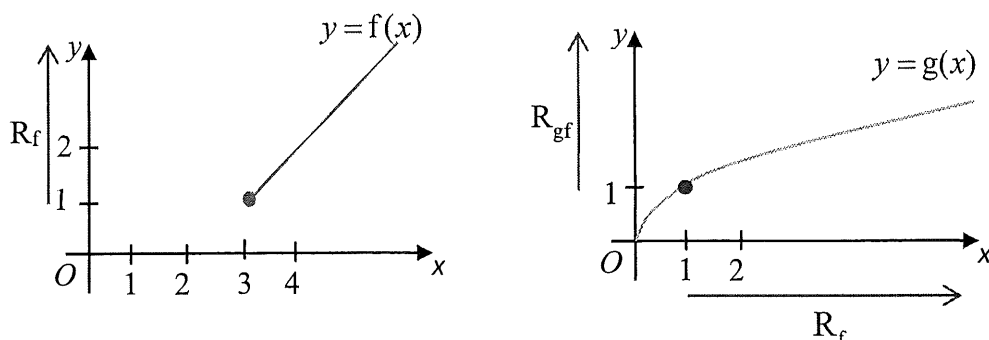
Find the range of the composite function gf given that the functions f and g are defined as follows:

$$\begin{aligned} f: x &\mapsto x-2, & x &\geq 3, \\ g: x &\mapsto \sqrt{x}, & x &\geq 0. \end{aligned}$$

Method 1

Sketch the graphs of the functions f and g on separate diagrams.

For the composite function gf , f is first applied to each $x \in D_f$ to find the image $f(x)$, and then the rule for g is applied to $f(x)$ to obtain the image $g(f(x))$. i.e. $gf(x) = g(f(x))$.



Consequently, we can obtain the range of gf as the range of g whose domain is restricted range of f , i.e. $[3, \infty) \xrightarrow{f} [1, \infty) \xrightarrow{g} [1, \infty)$.

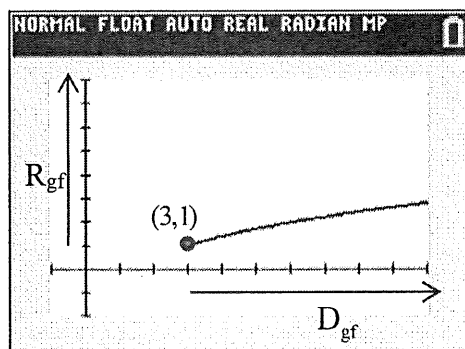
Thus $R_{gf} = [1, \infty)$.

Method 2

Find the function gf followed by sketching its graph to find the range.

$$gf(x) = g[f(x)] = \sqrt{x-2}, \quad D_{gf} = D_f = [3, \infty)$$

From sketch, $R_{gf} = [1, \infty)$.



Revision Tutorial Questions**Source of Question: EJC/Y5 Promo/2018/Q3**

- 1 The function f is defined by $f : x \mapsto \ln|1 - 2x|$, $x \in \mathbb{R}$, $x \neq \frac{1}{2}$.

(i) Explain why f does not have an inverse. [1]

The function g is now defined by $g : x \mapsto \ln|1 - 2x|$, $x > a$ where a is a real constant.

(ii) State the minimum value of a such that g^{-1} exists. [1]

(iii) Using the value of a stated in (ii), define g^{-1} in a similar form and state the range of g^{-1} . [4]

Source of Question: NJC/Y5 Promo/2018/Q10

- 2 The functions g and h are defined by

$$g : x \mapsto \frac{8x}{x^2 + 2}, \quad x \geq k,$$
$$h : x \mapsto \frac{1}{4}(x - 2)^2, \quad 0 \leq x \leq 4.$$

(i) Find the smallest exact value of k such that g^{-1} exists. [2]

Use $k = 2$ for the remainder of this question.

(ii) Find the exact range of g . [1]

(iii) Sketch the graphs of

$$y = g(x), \quad y = g^{-1}(x) \quad \text{and} \quad y = gg^{-1}(x)$$

on the same diagram.

Hence find the exact solution of $g^{-1}(x) = g(x)$. [6]

(iv) Show that hg exists and determine the range of hg . [3]

Source of Question: MI/Promo/PU2/2017/Q6

3 A function f is defined by $f(x) = 4 - (3 - x)^2$ for $x \in \mathbb{R}$.

- (i) Show that f^{-1} does not exist. [1]
- (ii) If the domain of f is restricted to $x \leq k$, state the largest exact value of k for which the function f^{-1} exists. [1]
- (iii) Using the domain defined in part (ii), find f^{-1} and its range. [3]
- (iv) Solve the equation $f(x) = f^{-1}(x)$ exactly. [3]

Source of Question: NYJC/Prelim/2017/01/Q7

4 The functions f and g are defined by

$$f: x \mapsto e^{-x^2}, \quad x \in \mathbb{R}, x < 0,$$

$$g: x \mapsto \frac{1}{x+3}, \quad x \in \mathbb{R}, x \neq -3.$$

- (i) Show that g^{-1} exists, and define g^{-1} in a similar form. [2]
- (ii) State the solution set for $g g^{-1}(x) = x$. [1]
- (iii) Explain why fg^{-1} does not exist. [1]

Let the function h be defined by

$$h: x \mapsto g(x), \quad x \in \mathbb{R}, x < k,$$

where k is a real constant.

- (iv) Given that fh^{-1} exists, state the maximum value of k . [1]
- (v) For the value of k found in (iv),
 - (a) find the exact range of fh^{-1} , [2]
 - (b) solve $h(x) = h^{-1}(x)$. [2]

Source of Question: ACJC/Y5 Promo/2017/Q11

- 5 (a) The function f is defined by

$$f : x \mapsto \left| e^{x^2} - 2 \right|, \quad x \in \mathbb{R}.$$

If the domain of f is restricted to $k \leq x \leq 0$,

- (i) state the least value of k for which the function f^{-1} exists, [1]

- (ii) find $f^{-1}(x)$, stating the domain of f^{-1} . [3]

- (b) The functions g and h are defined by

$$g : x \mapsto \begin{cases} \sin^{-1} x & \text{for } -1 < x \leq 1, \\ \pi \left(1 - \frac{x}{2} \right) & \text{for } 1 < x \leq 3, \end{cases}$$

$$h : x \mapsto 2x, \text{ for } -\frac{\sqrt{3}}{4} \leq x \leq 1.$$

- (i) Show that the composite function gh exists and find its range, giving all values in terms of π . [3]

It is given that $g(x+4) = g(x)$ for all real values of x .

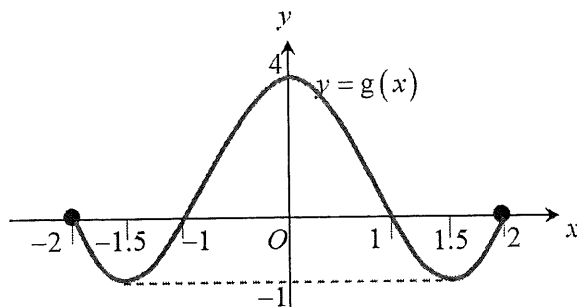
- (ii) Evaluate $g(17)$. [2]

- (iii) Sketch the graph of $y = g(x)$ for $-1 \leq x \leq 5.5$. [3]

Source of Question: TPJC/Prelim/2017/01/Q3

6 A function f is defined by $f : x \mapsto (x-k)^2$, $x < k$ where $k > 5$.

- (i) Find $f^{-1}(x)$ and state the domain of f^{-1} . [3]



The diagram above shows the curve with equation $y = g(x)$, where $-2 \leq x \leq 2$. The curve crosses the x -axis at $x = -2$, $x = -1$, $x = 1$ and $x = 2$, and has turning points at $(-1.5, -1)$, $(0, 4)$ and $(1.5, -1)$.

- (ii) Explain why the composite function fg exists. [2]
- (iii) Find in terms of k ,
 (a) the value of $fg(-1)$, [1]
 (b) range of fg . [2]

Source of Question: HCI/Promo/2018/Q6

7 The function f is defined by

$$f : x \mapsto \frac{x}{ax-1}, \text{ for } x \in \mathbb{R}, -\frac{1}{a} \leq x < \frac{1}{a},$$

where a is a positive constant.

- (i) Sketch the curve $y = f(x)$, stating the equation of the asymptote and the coordinates of the end-point. [2]
- (ii) Find $f^{-1}(x)$ and state the domain of f^{-1} . [3]
- (iii) Sketch the curve $y = f^{-1}(x)$ on the same diagram as the curve $y = f(x)$ in part (i). Hence obtain the range of values of x for which $f(x) = f^{-1}(x)$ in terms of a . [2]

The function g is defined by

$$g : x \mapsto \left(x - \frac{1}{3a}\right)^2 - \frac{1}{9a^2} \text{ for } x \in \mathbb{R}, x < \frac{1}{a}.$$

- (iv) Show that the composite function gf exists, and find the range of gf in terms of a .

[2]

Source of Question: SRJC JC2 Mid-Year CT 9758/2018/01/Q2

8 The functions f and g are defined by

$$f : x \mapsto 2 \cos \left(x + \frac{\pi}{4} \right), \quad -\pi \leq x \leq \pi,$$

$$g : x \mapsto (x-1)^2 + a, \quad -3 < x \leq 2 \text{ and } -1 < a < 0.$$

- (i) Explain why f does not have an inverse. [1]
- (ii) Explain why the composite function fg does not exist. Given that the composite function gf exists, find gf in similar form and find its range in terms of a . [4]
- (iii) The function f has an inverse if the domain is restricted to $-\pi \leq x \leq p$. State the largest possible exact value of p . [1]

Source of Question: SAJC/Prelim/2017/01/Q3

9 It is given that

$$f(x) = \begin{cases} b\sqrt{1 - \frac{x^2}{a^2}} & \text{for } -a < x \leq a \\ -a\sqrt{1 - \frac{(x-2a)^2}{a^2}} & \text{for } a < x \leq 3a \end{cases}$$

and that $f(x+4a) = f(x)$ for all real values of x , where a and b are real constants and $0 < a < b$.

- (i) Sketch the graph of $y = f(x)$ for $-a \leq x \leq 8a$. [3]
- (ii) Use the substitution $x = a \cos \theta$ to find the exact value of $\int_{3a}^{4a} f(x) \, dx$ in terms of a , b , and π . [5]

Source of Question: HCI/Prelim/2017/01/Q1

10 The *floor function*, denoted by $\lfloor x \rfloor$, is the greatest integer less than or equal to x . For example, $\lfloor -2.1 \rfloor = -3$ and $\lfloor 3.5 \rfloor = 3$.

The function f is defined by

$$f(x) = \begin{cases} \lfloor x \rfloor & \text{for } x \in \mathbb{R}, \quad -1 \leq x < 2, \\ 0 & \text{for } x \in \mathbb{R}, \quad 2 \leq x < 3. \end{cases}$$

It is given that $f(x) = f(x+4)$.

- (i) Find the values of $f(-1.2)$ and $f(3.6)$. [2]
- (ii) Sketch the graph of $y = f(x)$ for $-2 \leq x < 4$. [2]
- (iii) Hence evaluate $\int_{-2}^4 f(x) \, dx$. [1]